Computer Networks – Final Exam

Complete Question Set (40 Questions)

Part I: Multiple Choice Questions (1–40)

1. All of the following are the main features of SMTP, except ……
(A) Each object encapsulated in its response message.
(B) It uses persistent connections.
(C) Client push.
(D) It requires the message (header and body) to be in 7-bit ASCII.

2. What type of model does SMTP follow?
(A) Client-server model.
(B) Peer-to-peer model.
(C) Broadcast model.
(D) Hybrid model.

3. What is the role of the session layer in the OSI model?
(A) Mathematical formula shown
(B) Reliable delivery and end-to-end communication of data.
(C) Packet forwarding and routing of data packets.
(D) Establishing, managing, and terminating sessions or connections.
(E) Formatting, encoding, and compressing of data.

4. The UDP segment header consists of ……….. bytes.
(A) 24
(B) 16
(C) 32
(D) 20

5. Which network devices are typically involved in external networks?
(A) Servers and workstations.
(B) Routers and switches.
(C) Modems and gateways.
(D) Hubs and bridges.

6. In TCP, the sender will not overwhelm the receiver. It is called ……….
(A) Connection-oriented.
(B) Sequencing.
(C) Full duplex data.
(D) Flow control.

7. The UDP segment header contains the following information:
(A) Source IP address, Source port number, Destination IP address, Destination port number.
(B) Source port number, Destination port number, Seq. Number, Length.
(C) Source port number, Destination port number, Checksum, Length.
(D) Source IP address, Source port number, Destination IP address, Destination port number, Length, Checksum.

8. For the Internet/Guaranteed service model, the quality of service guarantees the following values for bandwidth, loss, order, and timing, respectively:
(A) possible, possibly, possibly, no.
(B) guaranteed min, no, yes, no.
(C) yes, yes, yes, yes.
(D) constant rate, yes, yes, yes.

9. ……… satisfy client requests without involving the origin server.
(A) Cookies.

(B) Sensors.
(C) Web caches.
(D) Databases.

10. ……… processes can be contacted.
(A) Client.
(B) P2P.
(C) Server.
(D) Internet.

11. Which layer handles addressing and routing in TCP/IP?
(A) Network interface layer.
(B) Transport layer.
(C) Internet layer.
(D) Application layer.

12. ……… protocol is responsible for routing, forwarding, addressing, and packet handling operations.
(A) OSPF.
(B) IP.
(C) BGP.
(D) ICMP.

13. ……… cable is two concentric conductors that can be used bidirectionally with multiple frequency channels on cable.
(A) Coaxial.
(B) Fiber optic.
(C) Twisted pair category 5.
(D) Twisted pair category 6.

14. ……… data and TV are transmitted at different frequencies over cable distribution network using ……….
(A) digital subscriber line, DSLAM.
(B) cable-based access, DSLAM.
(C) cable-based access, CMTS.
(D) digital subscriber line, CMTS.

15. ……… cable carries light pulses, each pulse a bit. It is also immune to electromagnetic noise.
(A) Fiber optic.
(B) Coaxial.
(C) Twisted pair category 6.
(D) Twisted pair category 8.

16. HTTP uses port number ………, whereas SMTP uses ……….
(A) 60, 35.
(B) 20, 60.
(C) 85, 90.
(D) 80, 25.

17. In the network layer, ……… places information in the datagram to control forwarding among routers along the end-to-end path from source to destination host.
(A) data.
(B) control.
(C) service.
(D) information.

18. Which protocol is used for retrieving emails from a mail server?
(A) SMTP.
(B) POP3.

(C) IMAP.
(D) FTP.

19. What does ATM stand for?
(A) Automated Teller Machine.
(B) Advanced Transmission Mode.
(C) All-Time-Money.
(D) Asynchronous Transfer Mode.

20. The following are reflections on best-effort services, except ……….
(A) The simplicity of the mechanism has allowed the Internet to be widely deployed and adopted.
(B) Insufficient quality mechanisms for the performance of real-time applications.
(C) Application layer distributed services located close to clients' networks allow services from multiple locations.
(D) Guaranteed delivery with less than 40 ms delay.

21. Which wireless standard provides the highest data transfer rate?
(A) 802.11b.
(B) 802.11g.
(C) 802.11n.
(D) 802.11ac.

22. In ……… the underlying channel may corrupt the packet. The receiver explicitly tells the sender which packets were received OK.
(A) Stop-and-wait.
(B) Go-Back-N.
(C) Selective Repeat.
(D) Pipelining.

23. Which standard is commonly used for Gigabit Ethernet over copper wire?
(A) 1000BaseX.
(B) 1000BaseT.
(C) 1000BaseF.
(D) 1000BaseSX.

24. Which protocol is used to determine the physical address of a host on the network?
(A) IMAP.
(B) ARP.
(C) ICMP.
(D) UDP.

25. ……… moves arriving packets from the router's input link to the appropriate router output link.
(A) Routing.
(B) Switching.
(C) Demultiplexing.
(D) Forwarding.

26. Which algorithm is commonly used in link-state routing protocols like OSPF?
(A) Distance-Vector algorithm.
(B) Floyd-Warshall algorithm.
(C) Bellman-Ford algorithm.
(D) Dijkstra's algorithm.

27. What is the purpose of a subnet mask?
(A) To represent IP addresses in binary format.
(B) To indicate the number of available host addresses in a network.
(C) To separate the network portion from the host portion of an IP address.
(D) To indicate the class of an IP address.

28. UDP is suitable for applications that prioritize ……….
(A) Low latency and real-time data transmission.
(B) Reliability.
(C) Flow control.
(D) Ordered data transmission.

29. Using ……… the host dynamically obtains the IP address from the network server when it joins the network.
(A) DHCP.
(B) OSPF.
(C) ARP.
(D) NAT.

30. TCP congestion and flow control are offered due to ………
(A) Flow control.
(B) Connection-oriented nature.
(C) Hybridization.
(D) Full duplex data.

31. ……… is a formal approach enabling feedback specifications, organizational rules and standards relevant to Internet and networking technologies, including routing, addressing and transport technologies.
(A) ISP.
(B) RFC.
(C) DSLAM.
(D) IETF.

32. What does MAN stand for?
(A) Multiple Area Network.
(B) Mobile Area Network.
(C) Metropolitan Area Network.

33. What is the purpose of the OUI in MAC addresses?
(A) To identify the IP address of the device
(B) To determine the network segment of the device
(C) To encrypt the data transmitted over the network
(D) To identify the manufacturer of the network equipment

34. What are the services provided by the data link layer?
(A) Framing, error detection and correction, flow control
(B) Logical addressing, routing
(C) Segmentation, reassembly, error checking
(D) Session establishment, maintenance, synchronization

35. Which routing approach is used in large networks with complex topologies?
(A) Dynamic routing
(B) Static routing
(C) Default routing
(D) Anycast routing

36. Which mode of operation involves the output of data from the sender being received by all other receivers?
(A) Simplex
(B) Broadcast
(C) Half Duplex
(D) Full Duplex

37. What is the purpose of a loopback address in IP addressing?
(A) To identify the network to which a device belongs
(B) To encrypt data transmitted over the network

(C) To assign unique IP addresses to devices
(D) To test network connectivity on a device

38. Which technology uses a fixed packet size of 53 bytes?
(A) Ethernet
(B) ATM
(C) DQDB
(D) Satellite systems

39. What is the reserved IP address class?
(A) Class E
(B) Class B
(C) Class A
(D) Class C

40. All of the following are DNS services, except ………
(A) Centralize hostname-to-IP-address translation database
(B) Host aliasing
(C) Mail server aliasing
(D) Load distribution

Part II: Long Answer Question

An ISP is granted a block of addresses starting with 190.100.0.0/16 (65536 addresses).

The ISP needs to distribute these addresses to three groups of customers as follows:

(a) The first group has 64 customers; each needs 256 addresses.

(b) The second group has 128 customers; each needs 128 addresses.

(c) The third group has 128 customers; each needs 64 addresses.

Design the subblocks and determine the number of addresses available after these allocations.